

Ume Taqadus

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Professional Summary

Final-year Computer Science student specializing in Artificial Intelligence, Machine Learning, and Generative AI, currently serving as Team Lead in the AI & Generative AI Summer Fellowship at Zeppelin Labs. Experienced in developing LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, AI agents, NLP pipelines, computer vision models, and AI automation workflows. Skilled in transformer fine-tuning (PEFT/LoRA), prompt engineering, vector databases, LLM orchestration, and deploying scalable AI solutions using PyTorch, TensorFlow, Hugging Face, LangChain, FastAPI, Docker, and cloud technologies.

Education

BS Computer Science

2022 – Present

FAST National University of Computer and Emerging Sciences

Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Science, Data Structures & Algorithms, Database Systems, Software Engineering, Cloud Computing

Technical Skills

Programming: Python, C++, C, JavaScript, SQL, Bash, Linux

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns, Clean Code, Debugging, Unit Testing

AI & Data Libraries: NumPy, Pandas, SciPy, Matplotlib, Scikit-learn

Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Regression, Classification, Clustering, XGBoost, LightGBM

Deep Learning: PyTorch, TensorFlow, Neural Networks, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms

Generative AI: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, Function Calling, MCP (Model Context Protocol), PEFT, LoRA, QLoRA, Fine-Tuning, Embeddings, Tokenization

LLM Frameworks: LangChain, LlamaIndex, Hugging Face Transformers, Ollama, OpenAI API, Gemini API, Anthropic Claude API

Natural Language Processing: Text Classification, Named Entity Recognition (NER), Semantic Search, Sentiment Analysis, spaCy, NLTK

Computer Vision: OpenCV, YOLO, Segment Anything (SAM), Image Classification, Object Detection, Image Segmentation

LLM Evaluation: RAG Evaluation, Prompt Evaluation, Model Comparison, Hallucination Detection, Response Quality Analysis

Frameworks & Backend: FastAPI, Flask, Django, Streamlit, React.js, Node.js, REST APIs, API Authentication

Databases: MySQL, PostgreSQL, MongoDB, Firebase, Vector Databases (FAISS, ChromaDB, Pinecone)

Cloud & Deployment: Docker, AWS (Basics), Google Cloud (Basics), Hugging Face Spaces, Model Deployment, Model Serving

MLOps: CI/CD, Experiment Tracking, Model Versioning, ML Pipelines, Deployment Workflows

AI Automation: n8n, Workflow Automation, AI Agents, API Integration

Networking: HTTP/HTTPS, OAuth, JWT, JSON, Client-Server Architecture

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

Experience

Team Lead – AI & Generative AI Summer Fellowship — Zeppelin Labs (Remote) Jul 2026 – Present

- Leading an AI engineering team to design and develop Generative AI applications using modern LLM architectures and collaborative software development practices.

- Architecting AI solutions involving Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI agents, vector databases, and intelligent automation workflows.
- Implementing prompt engineering strategies, LangChain, LlamaIndex, and embedding-based retrieval pipelines to improve AI system reliability and response quality.
- Managing Git-based development workflows including branching strategies, pull requests, code reviews, and team collaboration.
- Exploring and applying advanced Generative AI concepts including Agentic AI, Model Context Protocol (MCP), tool calling, workflow automation, and LLM orchestration.
- Mentoring team members, assigning technical tasks, reviewing implementations, and ensuring successful project delivery.

NLP Intern — Elevvo Pathways (Remote)

Aug 2025 – Sep 2025

- Developed NLP pipelines for text classification and Named Entity Recognition (NER) using transformer-based models.
- Fine-tuned transformer architectures using Hugging Face, improving evaluation performance by 10–15%.
- Applied prompt engineering and structured prompting techniques to improve LLM-generated responses.
- Performed data preprocessing, model evaluation, and experimentation for NLP-based AI solutions.

Junior AI/ML Developer — Code Generation (Remote)

Jul 2025 – Sep 2025

- Developed AI automation workflows integrating machine learning models, LLMs, and external APIs.
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines using FAISS vector search and embedding-based retrieval.
- Built LLM-powered automation modules to optimize business workflows and improve information processing.
- Worked with Python-based AI frameworks including TensorFlow, PyTorch, and Hugging Face.

AI/ML Intern — Arch Technologies (Remote)

Jun 2025 – Aug 2025

- Fine-tuned LLaMA-based language models using Parameter-Efficient Fine-Tuning (PEFT) and LoRA techniques.
- Developed a computer vision pipeline for medical image analysis using YOLO and segmentation models.
- Optimized model inference workflows to improve computational efficiency and reduce latency.
- Performed model training, evaluation, and experimentation using PyTorch-based deep learning frameworks.

Front-End Intern — TechNetCloud (Remote)

Jun 2024 – Sep 2024

- Developed responsive and interactive web interfaces using React.js, JavaScript, and Bootstrap.
- Improved application performance through component optimization, code splitting, and frontend best practices.
- Collaborated with developers to implement scalable and maintainable frontend solutions.

Projects

Lightweight Adaptive Routing System for LLMs and AI Agents

[GitHub](#)

- Designed an intelligent LLM routing framework that dynamically selects optimal models based on query complexity, performance, and computational cost.
- Developed an AI orchestration layer supporting multiple LLMs and specialized AI agents.
- Integrated Retrieval-Augmented Generation (RAG) pipelines for context-aware and reliable responses.
- Implemented explainability mechanisms to provide transparent routing decisions and model selection reasoning.
- Built an offline-first architecture optimized for lightweight and edge AI deployment.

AI Resume Screener & Feedback System

[GitHub](#)

- Developed an AI-powered ATS resume screening system that evaluates resumes against job descriptions using LLMs and semantic search.
- Implemented document processing pipelines for resume parsing, information extraction, and skill analysis.
- Used embeddings and vector databases for resume-job similarity matching and intelligent candidate ranking.
- Integrated LLM-based feedback generation to provide personalized resume improvement suggestions and skill-gap analysis.
- Designed an automated evaluation workflow combining NLP techniques, retrieval systems, and Generative AI.

University Assistant – RAG-Based Q&A System

[GitHub](#)

- Built an AI-powered university assistant using Retrieval-Augmented Generation over official university documents.
- Designed document ingestion, preprocessing, chunking, and embedding pipelines for semantic search.
- Integrated vector database retrieval with LLM generation to provide accurate, context-grounded answers.
- Reduced hallucinations by implementing retrieval-based response generation using verified university information.
- Developed an interactive question-answering interface for admissions, academic policies, and student procedures.

AI Email Manager and Action Assistant

[GitHub](#)

- Built an AI-powered email automation system using n8n, LLMs, and API integrations for intelligent inbox management.
- Implemented email classification into categories including Urgent, Important, Normal, Promotional, and Spam.
- Automated email workflows including filtering, summarization, prioritization, and action recommendation.
- Generated AI-powered summaries and recommended next steps for important communications.
- Designed workflow automation pipelines integrating AI services with productivity tools.

Offline Voice AI Assistant

[GitHub](#)

- Developed a fully offline voice AI assistant enabling privacy-preserving natural language interactions.
- Integrated Faster-Whisper for Speech-to-Text, local LLM inference for response generation, and Meta MMS-TTS for speech synthesis.
- Built an end-to-end voice pipeline supporting real-time speech recognition, reasoning, and voice responses.
- Optimized architecture for low-latency AI interaction on local hardware without cloud dependency.

Career Recommendation Chatbot

[GitHub](#)

- Developed an NLP-based career guidance chatbot using Hugging Face Transformers and Streamlit.
- Implemented intent classification and conversational workflows for personalized career recommendations.
- Applied NLP preprocessing and transformer-based models for user query understanding.

YOLO Tumor Detection System

[GitHub](#)

- Developed a computer vision pipeline for medical image analysis using YOLO and OpenCV.
- Implemented object detection and segmentation workflows for tumor localization.
- Applied image preprocessing, augmentation, and model evaluation techniques to improve robustness.

Certifications

- Claude Code in Action – Anthropic
- Claude Code 101 – Anthropic
- Artificial Intelligence – DigiSkills.pk
- Machine Learning – SAMZ Technologies

- Microservices with Docker – Udemy
- Google Carbon Programming

Achievements

- NSCT 2026: 75.5 percentile (HEC Pakistan national CS/AI benchmark exam)
- Selected as Team Lead for AI & Generative AI Summer Fellowship at Zeppelin Labs.

Soft Skills

Leadership — Team Management — Problem Solving — Critical Thinking — Technical Communication — Collaboration — Adaptability — Time Management — Research & Learning — Decision Making